

The path, not the end state: household fleet simulation, estimated behaviour and exploratory backcasting for robust and acceptable low-emission transport

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Working prototype and every number in this proposal: github.com/Astheno-sphere/computational_Logistics

why now

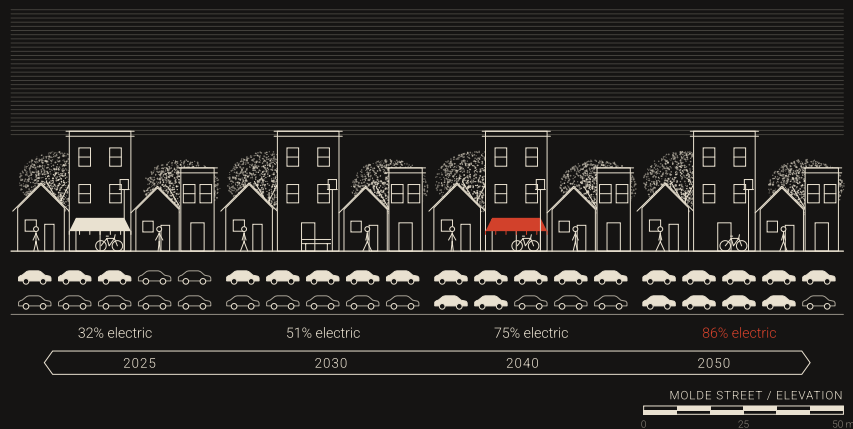
1 · THE ROAD, NOT THE DESTINATION

Norway won the showroom. The fleet, the tax base and the politics have not caught up.

In 2025, 95.9% of new cars sold in Norway were fully electric¹, yet most cars on the road still burn fuel, and a car stays in the fleet for close to two decades. The fuel tax that financed roads is eroding, and the distance-based charge meant to replace it has stalled between the state and the municipalities². The National Transport Plan 2025 to 2036 commits to better methods for steering toward the 2050 low-emission society³, and the Ministry's methods assignment admits that today's models suit continued policy, not deep cuts⁴. TRANSPLAN was created to work backwards from the 2050 goals⁵.

TØI's own review names the gap in one line: RTM and NTM6 compute partial equilibria but "say nothing about the path to a new equilibrium"⁶. **Backcasting is a path**, and that path runs through households: the car each one owns, the year it is replaced, the trips it makes and the charges it will accept. My PhD builds the engine that gives TRANSPLAN that path from people's estimated preferences, and tests whether the plans built on it survive futures nobody can forecast.

the stock problem, in one street



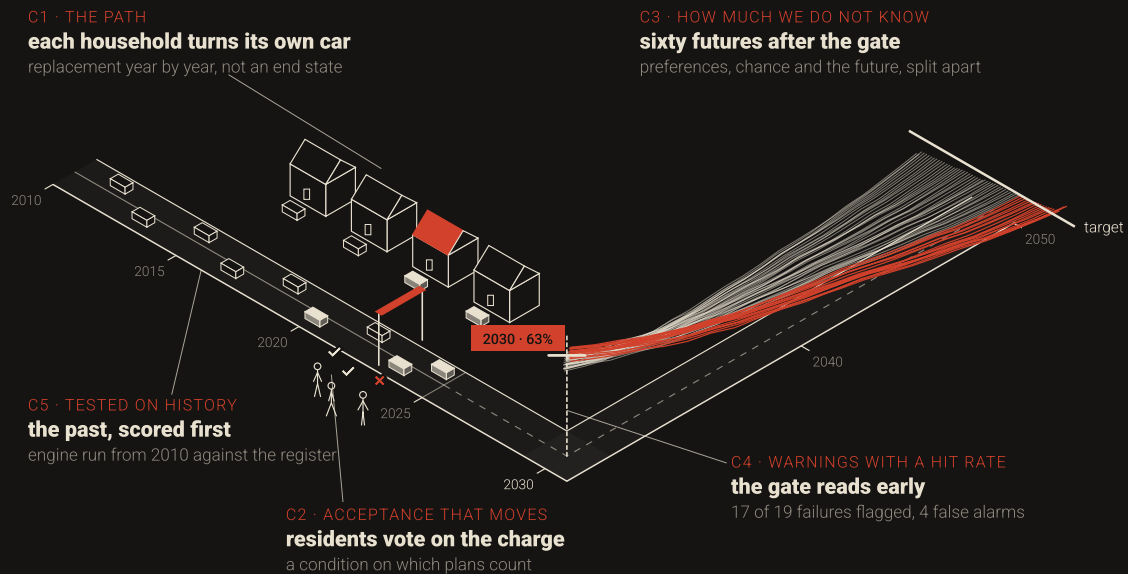
THE STOCK TURNS

one street, four moments. each lit car is electric, in the share the household engine simulates for that year under the most robust package. the street changes only as fast as households replace the cars they already own.

Prototype household engine, median future, package P12; without policy the 2050 share is 74%. Synthetic behaviour.

2 · FIVE CONTRIBUTIONS

the five contributions, on one road



FIVE QUESTIONS, ONE ROAD

time runs along the road. the engine is first scored on the record (C5). households turn the fleet one car at a time (C1) and vote on the charge they drive under (C2). at the 2030 gate one reading warns early (C4). after the gate the road opens into sixty futures, and a plan counts only if it holds in most of them (C3).

Fan: prototype ensemble, package P12, 60 futures, CO₂ relative to 2025; red futures miss the 2050 target. Synthetic behaviour.

C1 · THE PATH

A household engine that shows how Norway gets from today to 2050, year by year

Agents carry each household's car, its age, its replacement year, its trips and its response to price. Norway's fleet model BIG follows vintages with an estimated choice model, but in aggregate and for fixed scenarios⁷.

nearest: Fridstrøm et al. 2016; Deuten et al. 2020

C2 · ACCEPTANCE THAT MOVES

Acceptance estimated from choice data, changing as a charge is lived, and used as a condition on which plans count

Nationally, 70 to 80% support moderate distance charges with earmarked revenue, least among drivers with few alternatives⁸, and support rises once a charge is in place⁹. Le Pira, Marcucci and Gatta gave freight agents acceptance¹⁰; I carry it into passenger policy and into the robust search itself.

nearest: Le Pira et al. 2017; Lempert et al. 2020

C3 · HOW MUCH WE DO NOT KNOW

2050 uncertainty split into three sources, and a test of whether the winning plan survives each

Error in the estimated preferences, chance inside the simulation and deep uncertainty about the future, separated over 25 years. Petrik and colleagues propagated parameter error through one simulated day, with no long horizon and no policy choice¹¹.

nearest: Petrik et al. 2016, 2020; Tveter, Welde & Odeck 2025

C4 · WARNINGS WITH A HIT RATE

Pathways stress-tested across thousands of futures, with signposts whose lead time and false-alarm rate are measured

Measured signpost detection appears mainly in water management^{12,13}; transport backcasting remains national and deterministic¹⁴. Each signpost gets years of warning, false alarms and missed failures.

nearest: Raso et al. 2019; Haasnoot et al. 2018

C5 · TESTED ON HISTORY

The engine is checked against Norway's 2010 to 2025 electric transition before it is trusted with 2050

EV forecasts often fail against what happened¹⁵, and Norwegian fixed-link forecasts missed long-run traffic by up to a factor of two²⁴. The vehicle register lets the engine start from 2010 information and be scored against the observed fleet.

nearest: Deuten et al. 2020 (2010 to 2017)

where

3 · TWO TOWNS, ONE STREET, ONE BRIDGE

Molde and Kristiansund are **case studies, not a national sample**. They are car-dependent towns where national surveys find road charges hardest to accept, in a county whose climate accounts are off track¹⁶, and where tolls, bus service, parking and cycling are decided locally. Their networks concentrate travel: routing every pair of junctions by its fastest path, 17.5% of all routes in Molde use Øvre veg and a quarter of those in Kristiansund cross Gomabrua, so one charge or bus lane changes travel for most of the town at once. Molde is the main case, where preferences are estimated and the engine calibrated. Kristiansund is the transfer case: the Molde model is applied without re-estimation, which turns the transferability of preferences from an assumption into a measurement. Inputs are national and outputs are reported against national targets, so the method carries to other regions.

how

4 · ONE ENGINE, THREE ARTICLES

The engine has four layers: preferences estimated from people, the real road network, households carried through time, and the futures every plan is tested against. Each article builds one layer on the one below and hands a defined product to the next.

A1 · TRANSPLAN RA3 · YEAR 1

Preferences and acceptance

How do people in Molde and Kristiansund trade off time, cost and service, when do they replace their car, and which charging packages would they accept? Does national acceptance hold in car-dependent towns, and do preferences transfer between them?

Design. A stated-choice survey with three blocks: mode choice, timing and type of the next car, and votes on charging packages that vary charge level, exemptions, alternatives offered and use of revenue, with attribute ranges matched to the national experiment⁸. An efficient design is piloted on about 50 respondents; the full survey targets 800 respondents with eight tasks each, a size checked in advance by recovering all six prototype parameters within one standard error.

Recruitment. A Norwegian web panel (Norstat) combined with invitations through the two municipalities; assuming a 15% response, about 5,500 invitations are needed, a small share of two towns of 33,000 and 24,000. Hypothetical bias is reduced by a cheap-talk script, a consequentiality statement and levels pivoted on each respondent's own car and commute. **Models.** Mixed and latent-class logit, willingness to pay by segment, and acceptance with heterogeneity in trust and alternatives. **Product.** Preferences with their covariance and an acceptance function the agents carry.

A2 · TRANSPLAN RA2 · YEAR 2

The household fleet engine

When estimated preferences drive households with real cars, how do fleet renewal, mode shift, acceptance and who pays evolve year by year, and how much of the 2050 spread comes from preferences, from chance and from the future?

Engine. Households hold cars of known age and fuel, replace them by the A1 choice model, choose modes on the OpenStreetMap network, and form opinions on charges that shift with experience, calibrated on the Oslo field trial⁸; peer influence is bounded by sensitivity analysis, not claimed as identified. Preference parameters are drawn from their estimated distribution in every run. **Validation.** Calibration matches several observed patterns at once (travel survey mode shares, fleet age and fuel from the register, traffic counts)¹⁷, with Kristiansund held out and 2010 to 2025 as a historical test, documented with the ODD protocol¹⁸. **Scope and links.** Commuting and other regular car travel by residents, reported on that basis and translated to national targets. Emissions are well to wheel with Norwegian grid factors, rebound included; welfare is consumer surplus, comparable with Norwegian appraisal values; zone data come from RTM. The engine is built in year 1 on synthetic preferences, so year 2 swaps in real ones rather than starting from zero. **Product.** A calibrated engine and an ensemble interface.

A3 · TRANSPLAN RA1 · YEAR 3

Robust pathways and warnings

Across thousands of futures, which packages stay effective, acceptable and affordable, when do they fail, and which 2030 reading warns early, with what lead time and false-alarm rate?

Decision problem. Robustness $R(p)$ is the share of futures in which commuting emissions in 2050 fall to τ of the base year; the search returns $p^* = \operatorname{argmax} R(p)$ subject to acceptance $A(p) \geq \frac{1}{2}$ and public cost $C(p) \leq B$. I use a satisficing measure because the 2050 goal is itself a threshold to be met, not a quantity to maximise, and because it needs no probabilities over futures nobody can assign. τ is tested at 10, 20 and 30%, with regret and stability reported beside it, so

any ranking that depends on the metric is shown rather than hidden. **Method.** About 10⁴ futures anchored on the national strategies¹⁴ and 10² packages of the instruments debated in 2026: distance charging, toll rings in which electric cars pay, transit frequency and fares, cycling and fleet renewal. A many-objective robust search¹⁹, scenario discovery²⁰, Sobol indices and, if run time requires, a surrogate. Signposts are scored for lead time, false alarms and missed failures¹², and pathways for how often they must change²¹. **Freight test.** The same search is run for urban freight on the same network, with the route solver already in the toolkit, to show the method is not tied to commuters. **Product.** Robust, acceptable pathways with measured signposts.

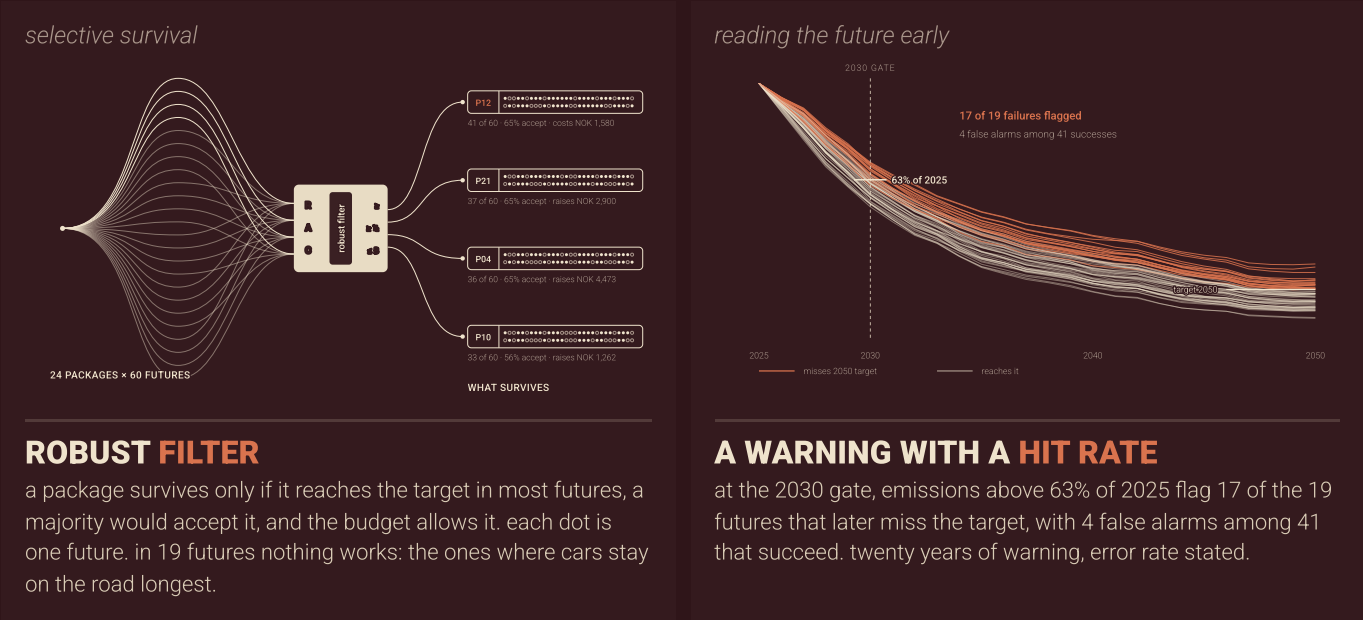
A4 · OPTIONAL

A workshop with county and municipal planners tests whether pathway stories help them agree on the next decision²².

proof

5 · IT ALREADY RUNS

A working prototype runs the full chain on synthetic behaviour and the real networks of both towns: two choice-model estimators with a recovery test, an agent model with fleet turnover, welfare and public money per household, and 1,500 runs of 25 packages in 60 futures. The behaviour is synthetic, so the numbers show what each article will produce, not what will happen in Norway. The code is public, 87 automated tests run on every change, and every number here is rebuilt by committed scripts.



Prototype ensemble, 1,500 runs; target: commuting CO₂ in 2050 at most 20% of 2025. Synthetic behaviour.

when

6 · PLAN, RISKS AND ETHICS

	YEAR 1	YEAR 2	YEAR 3
Research	Data applications in month 1; survey design, pilot, fieldwork; A1. Engine built in parallel on synthetic preferences	Real preferences swapped in; calibration, Kristiansund transfer, 2010 to 2025 test; A2	Ensemble, robust search, signposts, freight test; A3; thesis
Training	PhD courses: stated choice, econometrics, simulation	Research stay with a deep-uncertainty group	Optional A4 with planners through TRANSPLAN

Risks. Low response: web panel plus municipal channels, with the national travel survey as fallback. Delayed data: Statistics Norway commuting tables. Run time: surrogates and national computing. Weak identification: several observed patterns at once and a hold-out town. Results that hinge on the target or metric: reported across τ and metrics. **Ethics.** The survey is assessed by Sikt before fieldwork, with informed consent and pseudonymised storage; code is open under MIT, and data are released with a DOI where consent allows.

7 · FOR TRANSPLAN, THE REGION AND THE FIELD

For **TRANSPLAN**, a behavioural experiment for research area 3, the transition path national models lack for research area 2, and measured pathways for the backcasting of research area 1, built to sit alongside NTM6 and RTM and to move from “predict and provide” to “decide and provide”²³. For **Møre og Romsdal**, pathways with signposts planners can read from traffic counts and the vehicle register. For **the field**, acceptance as a moving condition on robustness, uncertainty split into its sources, and signposts with measured hit rates in transport.

The fleet that will drive in 2035 is being bought now. This project builds the instrument to steer it, and to know early when a plan needs to change.

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